Kisan Mitra

Smallholder Crop Advisory Agent

A photo and a spoken description become a cited diagnosis, a voice advisory in the farmer's own language, the nearest dealer with today's price, and a proactive outbreak alert - before the next farmer's crop even shows symptoms.

Theme: Sustainability & Social Impact

PROBLEM STATEMENT

The Problem

- A smallholder tomato farmer in Karnataka spots a diseased leaf with no agronomist to call and unreliable connectivity to search with.
- Existing diagnosis apps (Plantix, PlantVillage) identify the disease but stop there - they don't say where to buy the treatment nearby, what it costs, or whether a nearby outbreak means the farmer should check sooner.

TARGET USER

Who This Is For

- Smallholder tomato farmers in the Kolar tomato belt, Karnataka.
- Typically 0.5-4 acre plots, variable literacy, intermittent connectivity, no dedicated access to an agronomist.
- Primary language: Kannada.

SOLUTION OVERVIEW

What Kisan Mitra Does

- **1.** A cited disease diagnosis - not a black-box guess.
- **2.** A spoken advisory in the farmer's own language, weather-aware.
- **3.** The nearest dealer in stock for the recommended treatment, with today's price.
- **4.** A proactive outbreak alert when nearby plots report the same disease - before this farmer's own crop shows symptoms.
- **5.** When the crop is healthy and near harvest, the best nearby price to sell at today.
- **6.** A crop-loss insurance evidence packet, a group-buy discount alert, and an aggregated district dashboard for extension officers.

BEYOND DIAGNOSIS

From Advice to Economic Action

- **Insurance claim assistant:** a severe, confident diagnosis auto-generates a structured evidence packet (citation, confidence, estimated yield loss) formatted for a crop-loss insurance claim - most farmers with insurance never file because the paperwork is a barrier they can't clear alone.
- **Group-buy alert:** when an outbreak signal shows 3+ nearby farmers need the same treatment, checks whether combined demand clears the dealer's bulk-pricing threshold and surfaces a group discount - turning an early warning into coordinated economic action.
- **District dashboard:** an anonymized, village-level aggregate view of outbreak clusters for a local extension officer or NGO - not built for one farmer, built for deciding where to send resources district-wide.

ARCHITECTURE

How It's Built

- Farmer (photo + voice) -> Firebase Hosting -> Cloud Run (FastAPI backend)
- Gemini multimodal diagnosis, grounded via Vertex AI Search over the regional disease guide
- Cloud Translation + Text-to-Speech for the voice advisory
- Firestore holds the dealer directory, mandi prices, and diagnosis history
- Diagnosis history feeds three agentic reasoning steps: a geo-radius outbreak query across farmers, a per-plot history query for recurring issues, and a district-wide aggregation for the extension-officer dashboard
- An outbreak signal additionally triggers a bulk-pricing check against the dealer directory for the group-buy alert

How Generative AI Is Used

- **Gemini multimodal:** diagnoses from a photo plus a spoken description together, citing the specific passage that supports the diagnosis rather than guessing.
- **Vertex AI Search (RAG):** grounds every diagnosis in the actual regional extension guide, so the system can say why, not just what.
- **Gemini function calling:** orchestrates which tool to call next - dealer lookup when diseased, sell-price lookup when healthy and near harvest, group-buy check when an outbreak signal fires.
- **Gemini synthesis:** turns a cluster of raw nearby diagnosis records into a plain-language early-warning alert, and packages a severe diagnosis into a structured insurance evidence summary.

IMPACT

Why This Matters

- Each piece exists separately today - Plantix diagnoses, AgroStar connects to dealers, PlantVillage pairs AI with weather - but none combine diagnosis, voice, and market linkage into one flow.
- None reason across farmers or across time on the same plot.
- The outbreak alert reaches a farmer before their own crop shows damage - something no existing tool in this space does.
- No existing tool turns a diagnosis into a fileable insurance claim, a coordinated bulk-buy discount, or a district-level resourcing signal - moving the whole system from advice to economic action and policy support.

Why This Is Sustainability, Not Just Social Good

- **Targeted treatment, not blanket spraying:** a cited diagnosis recommends the specific input needed (e.g. copper oxychloride for early blight) instead of a farmer guessing and over-applying agrochemicals - reducing chemical runoff into the surrounding soil and water.
- **Weather-timed application:** the live weather check avoids recommending a spray right before rain washes it off - less wasted product, less chemical runoff from a wasted application.
- **Earlier detection means less crop loss:** every tomato lost to a late-caught disease is wasted water, soil nutrients, and land-use already invested in that crop - the outbreak alert catches disease before symptoms even appear on a given plot.
- **Plot-level memory enables root-cause fixes:** flagging a recurring disease as a drainage or soil issue, not just re-treating the symptom, shifts the farmer toward a lasting fix instead of repeat chemical use season after season.

SCALABILITY & ROADMAP

What's Next

- **Built:** sell-side market linkage - extends the agent from "where to buy treatment" to "where to sell the harvest."
- **Planned:** an offline-first fallback - a distilled on-device classifier for the 8-10 most common regional diseases, so core diagnosis keeps working with no connectivity.
- **Expansion path:** a new crop or region means indexing a new disease guide into Vertex AI Search, not rewriting the diagnosis logic - crop-agnostic by design.

SUBMISSION

Built On Google Cloud

- **Gen AI:** Gemini (multimodal diagnosis, function calling, synthesis)
- **Retrieval:** Vertex AI Search
- **Deployment:** Cloud Run (backend) + Firebase Hosting (frontend)
- **Data:** Firestore (dealers, mandi prices, diagnosis history)
- **Accessibility:** Cloud Translation + Cloud Text-to-Speech